

Syllabus for (Graduate) Combinatorics I

Fall 2026, Howard University

Course #: Math 273
Class time: Mon-Wed, 11:10am-12:30pm
Classroom: Annex III (Graduate School) - #224
Website: <http://www.samuelhopkins.com/classes/273.html>

Instructor: Sam Hopkins
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Office: Annex III (Graduate School) - #220
Office Hours: By appointment

Course content: We'll study basic combinatorial objects (subsets, multisets, permutations, set/number partitions, compositions, graphs, trees, etc), their enumeration and additional structures they carry (such as partial orders).

Prerequisites: Calculus, linear algebra, undergraduate algebra (groups, rings, fields)
Textbook: R.P. Stanley, *Enumerative combinatorics*, Vol. I, 2nd ed.
F. Ardila, *Algebraic & geometric methods in enumerative combinatorics*
Problems come from Stanley, but the lectures more closely follow Ardila.

Grading: **50% 4 Homeworks | 25% Midterm Exam | 25% Final Exam**
There will be 4 homework assignments for the semester.
There will be one in person midterm exam, about halfway through.
There will be an in person final exam, during finals week.
Collaboration on the homework is encouraged, as long as each person understands the solutions, writes them up in their own words, and indicates with whom they collaborated. Collaboration on the exams is not allowed. Beyond that, I expect you to show up to class and be engaged

Detailed outline: Roughly speaking we will try to cover the following:

- Generating functions (ordinary and exponential)
- Basic objects:
 - integer partitions and compositions,
 - subsets and multisets,
 - permutations,
 - set partitions,
 - the Catalan families
- The 12-fold way
- Combinatorial statistics and q-analogs:
 - q-binomial and q-multinomial coefficients,
 - permutation statistics (Mahonian and Eulerian)
- Counting with signs:
 - the principle of inclusion-exclusion,
 - sign-reversing involutions

- Determinantal/trace formulas:
 - matrix-tree theorem,
 - transfer matrix method
- Partially ordered sets and lattices:
 - distributive lattices, Birkhoff's Theorem,
 - Möbius functions and Möbius inversion

But if there are any topics you are especially interested in, please let me know! I am happy to tailor this course to the interests of the students.

Reproduced below this line are various department- and school-wide policy statements.

Academic Code of Student Conduct (*please see Howard University handbook*): No copying, unauthorized use of calculators, books, or other materials, or changing of answers or other academic dishonesty will be tolerated. Cheating will not be tolerated. Anyone caught cheating will receive an F for the course and may be expelled from the university.

Grievance Procedure: If you have any problems with the policies or rules of this course, discuss your concerns with your instructor. If you are still unable to come to a satisfactory arrangement, you may contact the Director of Graduate Studies, Dr. Davenport, dennis.davenport@Howard.edu, and then, finally, the Chair of the Department, Dr. Toni, bourama.toni@howard.edu. See also the webpage: <https://mathematics.howard.edu/math-support-resources/student-grievance-procedure>.

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